Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C091 **VPH1** 3 genes / 2 species 2 direct pairs	**Y1 VPH1** GenAge 1719 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 18340043	**W1 unc-32** GenAge 705 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 10747056 **W2 vha-6** GenAge 715 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17411345	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); W2–Y1 (n:m)				
C011 **vps-33.1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 vps-33.1** GenAge 720 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	**F1 car** GenAge 89 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 17435236	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				
C096 **VPS20** 2 genes / 2 species 1 direct pair	**Y1 VPS20** GenAge 1745 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 22912585	**W1 vps-20** GenAge 764 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18006689	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C107 **VPS21** 2 genes / 2 species 1 direct pair	**Y1 VPS21** GenAge 1180 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20657825	**W1 rab-5** GenAge 975 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 19627265, 23144747	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.